

Prediction-guided mixed-model GWAS refines maize oil-trait candidate intervals in ZEAMAP

Title Page

Title: Prediction-guided mixed-model GWAS refines maize oil-trait candidate intervals in ZEAMAP

Running title: ZEAMAP oil-trait prediction and GWAS

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Core Ideas

- ZEAMAP processed data were harmonized into a conservative accession-level analysis set.
- Oil and fatty-acid traits were more predictable than other trait families under regularized genotype-based models.
- Mixed-model GWAS was essential: covariate-only testing was inflated, whereas GEMMA controlled genomic inflation.
- Recurrent chr6 and chr9 intervals provide the clearest fatty-acid candidate signals.
- The study prioritizes candidate intervals for follow-up and does not claim causal variants or experimentally validated genes.

Abstract

Maize kernel oil content and fatty-acid composition are important seed-quality traits, but association analyses in diverse maize panels must distinguish true local signals from population structure and relatedness. We harmonized public ZEAMAP processed data into a conservative accession-level analysis set containing 461 accessions, 199,856 filtered SNPs and 318 numeric phenotype or metabolite traits. Repeated genotype-to-phenotype benchmarking identified 66 robustly predictable traits; oil traits formed the strongest family, with a genotype plus population ridge model reaching median Pearson/R² of 0.597/0.338 across robust oil traits. This prediction result motivated focused GWAS for 10 high-priority oil traits. A covariate-only scan was strongly inflated, whereas GEMMA mixed linear models with genotype-derived kinship controlled lambda GC to 0.984-1.018 (median 0.998). We annotated and ranked 184 candidate loci and 147 candidate genes after excluding nominal-only signals. The strongest evidence converged on a recurrent chromosome 6 linoleic acid1-region interval and a chromosome 9 C16:0-associated interval containing nearby acyl-ACP thioesterase annotation. These results provide a reproducible framework for moving from public maize processed data to calibrated oil-trait candidate intervals while keeping causal claims appropriately bounded.

Keywords

maize; ZEAMAP; kernel oil; fatty-acid composition; genomic prediction; mixed-model GWAS; GEMMA; candidate interval

Introduction

Kernel oil concentration and fatty-acid composition influence maize grain quality, nutritional value and industrial use. The traits are also biologically informative because they connect quantitative genetic variation with fatty-acid metabolism, seed development and carbon allocation. Previous maize studies

have identified oil-content and fatty-acid loci, including signals related to linoleic-acid composition and DGAT-mediated oil accumulation (Alrefai et al., 1995; Zheng et al., 2008; Li et al., 2013; Zhang et al., 2023). Yet public diversity-panel data still require careful handling before these biological signals can be interpreted.

ZEAMAP provides a broad public resource for maize multi-omics and trait data (Gui et al., 2020). The practical challenge is that processed tables do not automatically form a single analysis matrix. Accessions, modalities and biological units must be reconciled before modelling. In particular, reference or tissue-level expression resources cannot be treated as accession-level expression measurements, and partially paired methylation data should not be used as if they covered the full panel. We therefore built a restrained accession-level analysis set in which genotype, population information and phenotype/metabolite traits were aligned before modelling.

The second challenge is statistical. A panel of 461 accessions and nearly 200,000 SNPs is large enough for genome-wide association and regularized prediction, but not large enough to justify unconstrained high-capacity prediction models. A model can appear sophisticated while learning population structure or noise. We therefore used regularized models and repeated evaluation to identify trait families with stable predictive signal, then used that result to prioritize a focused GWAS layer.

The third challenge is association calibration. Diversity-panel GWAS is vulnerable to residual structure. A candidate-gene story built on inflated tests is weak even when the gene annotation is appealing. For this reason we explicitly treated covariate-only GWAS as a diagnostic layer and used GEMMA mixed linear models as the manuscript-level association test (Zhou & Stephens, 2012; Zhou & Stephens, 2014). Lead loci were then interpreted through a conservative evidence hierarchy combining statistical support, recurrence across oil traits, prediction-overlap information and local gene annotation from B73 reference resources (Jiao et al., 2017; Yates et al., 2022).

Results

Accession harmonization produced a restrained ZEAMAP analysis set

After harmonizing processed ZEAMAP genotype, population and phenotype/metabolite tables, the analysis set retained 461 accessions with the required paired information, 199,856 filtered biallelic SNPs and 318 numeric traits. Methylation coverage was available for 236 accessions and was tracked as a partially paired modality rather than merged into the primary model. Expression resources available in the working data were not accession-matched to the diversity panel and were therefore excluded from genotype-to-phenotype prediction.

This filtering reduced the apparent size of the resource but increased interpretability. Every primary prediction and GWAS result was based on the same accession-level unit. This matters because oil-trait signals in a structured maize panel can be confounded if sample identity or modality pairing is loose.

Oil traits showed the strongest and most stable prediction signal

Repeated prediction benchmarking showed that oil traits were the clearest target family. Across selected trait families, oil traits had the highest median Pearson correlation (0.597) and median R² (0.338), with the best oil trait reaching Pearson/R² of 0.924/0.827. Among the robust final traits, 29 were oil-related. The best-performing traits included total oil content and major fatty-acid components, indicating that the genotype and population features captured biologically meaningful variation rather than only broad agronomic structure.

Model comparison also supported a conservative modelling choice. Ridge and ElasticNet were more stable than a small multilayer perceptron under the available sample size. The result is not a claim against neural networks in maize genetics; it shows that, for this data scale and pairing structure, regularized models provide a stronger basis for trait prioritization and biological follow-up.

Methylation was informative only as an auxiliary layer

Methylation was evaluated in three forms: global accession summaries, gene/promoter/cis-window principal components and sparse gene-window features. The global features showed little broad improvement, gene-level components provided limited auxiliary signal and sparse methylation features were unstable in the smaller 236-accession subset. We therefore did not include methylation in the primary prediction model. This decision prevents a weak multi-omics claim and keeps the main analysis anchored in the fully paired genotype and trait data.

GEMMA controlled inflation that persisted in covariate-only GWAS

Prediction results directed the association analysis toward 10 high-priority oil traits. The covariate-only scan was useful as a warning: genomic inflation remained high despite population covariates. In contrast, GEMMA mixed linear models using genotype-derived kinship controlled lambda GC to 0.984-1.018, with a median of 0.998. This calibration is central to the study because it supports interpretation of local peaks as candidate intervals rather than artifacts of residual relatedness.

Across the 10 oil traits, GEMMA tested 440 non-missing accessions and 199,856 SNPs per trait. Bonferroni-significant hits ranged from 1 to 21 per trait, with 126 Bonferroni-level hits and 338 FDR-level hits in total. The strongest signals occurred for very-long-chain fatty-acid composition, C18 fatty-acid traits and C16:0. Trait-level Manhattan and QQ plots were retained as diagnostic figures, while the main text focuses on calibrated summary statistics and prioritized intervals.

Candidate-locus ranking separated direct lipid candidates from broader regulatory regions

GEMMA lead signals were merged into physical candidate intervals and annotated with B73 RefGen_v4 gene models. After removing nominal-only signals, the candidate set contained 184 loci and 147 candidate genes. Of these loci, 63 also overlapped ridge-attribution evidence and 11 carried fatty-acid or lipid-related annotation. This layered ranking separated two types of evidence: direct lipid-metabolism candidates and recurrent but functionally broader regions.

The strongest interval was R01_Zm00001d036982 on chromosome 6. It was recurrent across seven oil traits, reached best $P = 2.35e-25$ and contained the B73 RefGen_v4 gene Zm00001d036982, which maps to the current B73 v5 gene Zm00001eb277490. Local annotation and external xrefs support lipid acyltransferase/DGAT-like or linoleic-acid biology. This region therefore has the strongest combination of recurrence, statistical support, prediction overlap and prior fatty-acid evidence.

The chr9 C16:0 interval provides a different but biologically important signal. The lead interval around Zm00001d045383 reached best $P = 7.76e-17$ for C16:0. The lead gene maps to a DXS/isoprenoid-related annotation, while the nearby Zm00001d045387/Zm00001eb377350 gene carries acyl-ACP hydrolase, palmitoyl-ACP thioesterase and fatty-acid biosynthesis evidence. This makes the region a high-priority saturated fatty-acid candidate interval, but not yet a fine-mapped FatB causal gene.

Other tier-1 intervals on chromosomes 1, 4 and 8 were recurrent and ridge-supported but had broader annotations, including MYB-domain, protein-transport and tetratricopeptide-repeat genes. These regions are valuable because recurrence across oil traits suggests reproducible association structure. However, their current annotations support candidate-interval reporting rather than direct oil-biosynthetic claims.

Expanded figure and table package clarifies the evidence hierarchy

The revised figure set contains four main figures. Figure 1 summarizes data harmonization and prediction benchmarking. Figure 2 presents the mixed-model GWAS calibration and candidate-locus summary. Figure 3 shows the chr6 and chr9 regional intervals. Figure 4 integrates trait-family predictability, GEMMA calibration, top-region association strength and evidence classes across prioritized loci. The table set contains one main evidence table for the eight top regional loci and four supplementary tables covering the full candidate-locus set, tier-1 loci, external annotation hardening and column definitions. This structure keeps the main text readable while giving reviewers enough information to inspect the full locus-ranking logic.

Discussion

This study provides a calibrated route from public ZEAMAP processed data to maize oil-trait candidate intervals. The most important result is not simply that several oil-trait associations were detected. Rather, the analysis shows why these associations are interpretable: the input accessions were harmonized, oil traits were prioritized by repeated prediction, covariate-only GWAS was shown to be inflated and GEMMA mixed models brought genomic inflation close to one.

The prediction results help explain why the GWAS focus on oil traits is justified. Oil and fatty-acid traits were not chosen only because they are biologically attractive. They were the most consistently predictable trait family in the analysis set, with strong performance for total oil and major fatty-acid components. This suggests that the available SNP features capture a meaningful part of the genetic architecture for these traits. In practical breeding terms, the result supports oil traits as a near-term target for regularized genomic prediction in this ZEAMAP-derived panel, while more weakly predictable metabolite or amino-acid traits may require larger sample sizes, better environmental metadata or different feature representations.

The contrast between covariate-only and mixed-model GWAS is equally important. In a structured maize panel, population covariates alone were not sufficient. Without the kinship term, the association scan produced inflation severe enough to undermine candidate interpretation. GEMMA did not merely improve a technical diagnostic; it changed the credibility of the biological conclusions. The near-one lambda values across all 10 oil traits indicate that the final candidate intervals are less likely to be dominated by residual relatedness.

The chr6 interval is the main biological anchor of the manuscript. Its strength comes from convergence: multiple oil traits point to the same region, the association is highly significant, ridge-derived evidence supports the region and local annotation connects it to linoleic-acid or lipid-acyltransferase biology. Previous maize oil and fatty-acid studies provide a plausible context for this signal (Alrefai et al., 1995; Zheng et al., 2008; Li et al., 2013; Zhang et al., 2023). The appropriate claim is therefore strong but bounded: this is the leading recurrent fatty-acid candidate interval in the present analysis, not a validated causal gene.

The chr9 C16:0 interval is more trait-specific and illustrates why gene-level interpretation must be careful. The lead gene itself is not annotated as FatB. The stronger fatty-acid interpretation comes from a nearby acyl-ACP thioesterase candidate, a pathway class directly related to saturated fatty-acid composition. This supports follow-up of the interval, especially for C16:0, but the result should not be overstated as a confirmed causal FatB allele.

The broader set of recurrent loci may be useful for breeding and functional prioritization. Some intervals contain regulatory or transport-related candidates rather than canonical oil-pathway genes. Such genes could still affect seed composition through developmental timing, transport, organelle function or resource allocation. However, these hypotheses need external expression evidence, fine mapping or functional assays. The revised evidence hierarchy makes this distinction explicit: lipid-annotated intervals are discussed as stronger candidates, while broader recurrent regions are retained as hypothesis-generating intervals.

Several limitations define the next experiments. First, the analysis set remains modest for high-dimensional prediction, so complex models should wait for larger paired datasets. Second, methylation was only partially paired and cannot yet support a strong multi-omics mechanism. Third, the GWAS resolution is limited by LD, marker density and accession sample size. Fourth, candidate genes were not experimentally validated. Stronger follow-up would include independent panels, seed-specific expression evidence, haplotype analysis around chr6 and chr9, and functional tests for the leading lipid-related genes.

Overall, the study is best positioned as a plant genomics resource and candidate-prioritization paper. It contributes a reproducible accession-level analysis, a statistically calibrated oil-trait GWAS layer and a ranked set of candidate intervals for maize oil and fatty-acid composition. The claims are intentionally conservative, which should make the manuscript more defensible in review.

Materials and Methods

ZEAMAP data and accession matching

Processed ZEAMAP public data were obtained from CNGBdb project CNP0001565 (Gui et al., 2020). Genotype, population, phenotype and metabolome tables were matched at accession level. Accessions were retained when genotype, population information and at least one phenotype or metabolome measurement were present. SNPs were filtered to common biallelic markers with sufficient sample coverage, yielding 199,856 variants for modelling and association analysis.

Prediction analysis

Genotype-to-phenotype prediction was evaluated using repeated train/validation/test splits. Candidate models included genotype plus population ridge regression, genotype plus population ElasticNet, population-only ridge regression and a small multilayer perceptron. Traits were retained for final reporting only when performance was positive and stable across seeds. Pearson correlation and R² on held-out samples were used as the main performance metrics.

Methylation assessment

Methylation information was evaluated as an auxiliary modality in the subset of accessions with coverage. We tested global methylation summaries, gene/promoter/cis-window components and sparse gene-window features. Because the methylation subset was substantially smaller than the full analysis set and sparse features were unstable, methylation was not used in the primary prediction model.

Mixed-model GWAS

Ten high-priority oil traits were selected from the robust prediction results. Covariate-only GWAS was used as a diagnostic comparison. The primary association analysis used GEMMA mixed linear models with genotype-derived kinship and population covariates (Zhou & Stephens, 2012; Zhou & Stephens, 2014). Likelihood-ratio p-values were used for locus ranking. The Bonferroni threshold was 0.05 divided by 199,856 tested variants; FDR summaries were used for candidate triage rather than causal claims (Benjamini & Hochberg, 1995).

Candidate-locus annotation and ranking

Lead SNPs were merged into physical intervals and annotated with B73 RefGen_v4 gene models (Jiao et al., 2017). Candidate intervals were ranked by association strength, recurrence across oil traits, overlap with prediction-derived evidence, local gene annotation and external gene-model support from plant genome resources (Yates et al., 2022). Intervals were classified as direct lipid candidates, indirect regulatory/transport candidates or unresolved recurrent candidates. Candidate genes were treated as hypotheses unless supported by independent functional evidence.

Software and reproducibility

Analyses used Python scientific-computing libraries and GEMMA for mixed-model association (Pedregosa et al., 2011; Harris et al., 2020; Virtanen et al., 2020; McKinney, 2010; Hunter, 2007; Zhou & Stephens, 2012). The repository includes the scripts and summary tables needed to reproduce the accession harmonization, prediction benchmark, GWAS summaries, candidate-locus ranking and manuscript figures.

Data Availability Statement

The study reanalyzes public ZEAMAP processed data from CNGBdb project CNP0001565. The project repository contains analysis scripts, summary tables, candidate-locus annotations and generated figures. Large public raw inputs and large intermediate matrices are not included in the repository; their source and regeneration paths are documented with the analysis files.

Code Availability Statement

Code used for dataset construction, prediction benchmarking, methylation assessment, GWAS summarization, candidate-locus annotation and figure preparation is available in the project repository under `scripts/`.

Author Contributions

xxx

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xxx

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xxx

Competing Interests

xxx

Ethics Statement

No human participants or animal subjects were used. The study reanalyzes public plant genomics data.

Figure Legends

Figure 1. ZEAMAP accession harmonization and prediction benchmark. The figure summarizes accession matching, retained data scale, model comparison and trait-family prediction performance.

Figure 2. GEMMA mixed-model GWAS calibration and candidate-locus summary. The figure shows that GEMMA controlled genomic inflation and summarizes the filtered candidate-locus set.

Figure 3. Regional evidence for chr6 and chr9 oil-trait candidate intervals. Panels highlight the recurrent chr6 linoleic acid1-region interval and the chr9 C16:0-associated acyl-ACP thioesterase candidate interval.

Figure 4. Integrated prediction and GWAS evidence. Panels summarize trait-family predictability, GEMMA calibration across oil traits, association strength of top regional loci and evidence classes across prioritized regions.

Table Legends

Table 1. Prioritized regional candidate loci for maize oil traits. Eight top loci are summarized by recurrence, association strength, prediction overlap, functional annotation and claim boundary.

Supplementary Table 1. Manuscript candidate loci from oil-trait GEMMA mixed-model GWAS.

Supplementary Table 2. Tier-1 candidate loci used for main-text interpretation.

Supplementary Table 3. External annotation review for top regional loci.

Supplementary Table 4. Column dictionary for supplementary tables.

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